

LCD MODULE SPECIFICATION

Model:	UE043HV-RN40-A058B
Version:	V1.1
Date:	20220516

Customer Confirmation 客户确认

Approved by	Notes

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Prepared by	Reviewed by	Approved by

REVISION HISTORY

Revision 版本号	Date 日期	Contents of Revision Change 修改内容	Remark 备注
V1.0	20220515	Preliminary release	
V1.1	20220516	Update IC	

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1. GENERAL INFORMATION

1.1 Features

- 1) Pixel Arrangement: RGB Vertical Stripe
- 2) Interface Mode: RGB 24bits
- 3) Driver IC: ILI6485A-M0T0061 CTP IC: GT911
- 4) Operation Temperature: -20~70℃
- 5) Storage Temperature: -30~80℃
- 6) Backlight Type: White LED
- 7) Display mode: Normally White,
- 8) Pixel Density: 128 PPI
- 9) LED life time: 30,000 Hours

1.2 Mechanical Specification

Item 项目	Specification 规格	Unit 单位	Remark 备注
Pixel Driving element	TN TFT	-	-
Screen Size	4.3	Inch	Diagonal
Resolution	480(W)*3(RGB)*272(H)	Dots	-
Interface	RGB 24bits	-	40PIN
Module Power Consumption	0.838	Watt	Typ.
Active Area	95.04(W)*53.86(H)	mm	-
Pixel pitch (W*H)	0.066(W)*0.198(H)	mm	-
CTP Module Size (W*H*D)	105.5(W)*67.2(H)*4.45(D)	mm	-
Luminance	250	cd/m ²	Typ.
Viewing Direction	12	O'clock	-
Display Color	16.7M	Colors	24bits

2. ABSOLUTE MAXIMUM RATINGS

Item 项目	Symbol 符号	Min. 最小值	Max. 最大值	Unit 单位	Remark 备注
Power supply voltage	VDD	-0.3	4.6	V	Note1
LED Reverse Voltage	V _R	-	5	V	For each led, Note1
Operating temperature	T _{op}	-20	70	°C	Note1,2
Storage temperature	T _{st}	-30	80	°C	Note1,2
Humidity	H _{st}	10	90	%RH	Note1,3

(Ta=+25°C, GND=0V)

Note1: If the module exceeds the absolute maximum ratings, it may be damaged permanently. Also if the module operates with the absolute maximum ratings for a long time, the reliability may drop.

Note2: In case of temperature below 0°C, the response time of liquid crystal (LC) becomes slower and the color of panel darker than normal one.

Note3: Temp. ≤ 60°C , 90% RH MAX.

Temp. > 60°C , Absolute humidity shall be less than 90% RH.

REV.	CONTENT	DESIGN	DATE
A	FIRST RELEASE	肖春光	2022.0120
B	换C,改PC尺寸	刘春发	2022.0516

PN	NOTE
1	RST
2	VCC 3.3V
3	GND
4	INT
5	SDA 3.3V
6	SCL 3.3V

NO	PIN	NAME
1	1	LEAK
2	2	LEAK
3	3	LEAK
4	4	VDD
5	5	R0
6	6	R1
7	7	R2
8	8	R3
9	9	R4
10	10	R5
11	11	R6
12	12	R7
13	13	G0
14	14	G1
15	15	G2
16	16	G3
17	17	G4
18	18	G5
19	19	G6
20	20	G7
21	21	R0
22	22	B1
23	23	B2
24	24	B3
25	25	B4
26	26	B5
27	27	B6
28	28	B7
29	29	GND
30	30	CLK
31	31	DISP
32	32	HSYNC
33	33	VSYNC
34	34	DEN
35	35	GND
36	36	GND
37	37	AR
38	38	Y
39	39	X
40	40	YU

4. I/O CONNECTION & BLOCK DIAGRAM

4.1 I/O Connection

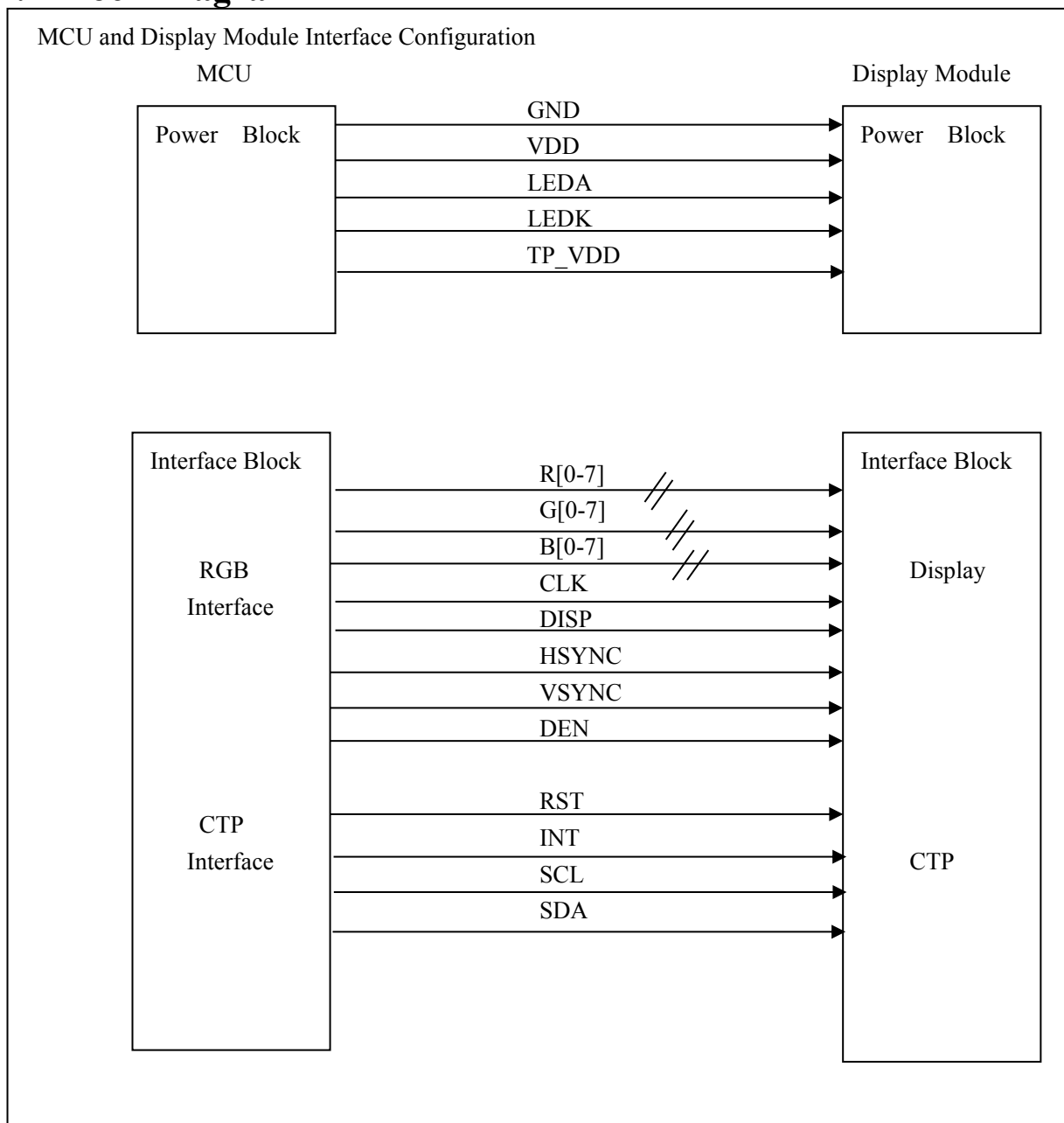
Pin No. 序号	Symbol 符号	I/O	Description 描述
1	LEDK	P	Power supply for backlight cathode
2	LEDA	P	Power supply for backlight anode
3	GND	P	Power Ground
4	VDD	P	Power supply to the internal logic power regulator(3.3V)
5-12	R0-R7	I	Red data input.
13-20	G0-G7	I	Green data input.
21-28	B0-B7	I	Blue data input.
29	GND	P	Power Ground
30	CLK	I	Pixel clock input pin, Negative polarity
31	DISP	I	Standby mode. Normally pulled high.
32	HSYNC	I	Horizontal sync signal, Negative polarity
33	VSYNC	I	Vertical sync signal, Negative polarity
34	DEN	I	Data input enable. Display access is enabled when DE is “H”
35	NC	I	Dummy
36	GND	P	Power Ground
37	XR	-	Dummy
38	YD	-	Dummy
39	XL	-	Dummy
40	YU	-	Dummy

I: Input; O: Output; P: Power

TP

Pin No. 序号	Symbol 符号	I/O	Description 描述
1	RST	P	The signal will reset the TP,Signal is active low
2	VCC3.3	I	Power supply for TP
3	GND	O	Power Ground
4	INT	I	Interrupt signals for TP
5	SDA 3.3V	I/O	I2C data signal for TP
6	SCL 3.3V	P	I2C clock signals for TP

4.2 Block Diagram



5. ELECTRICAL CHARACTERISTICS

5.1 TFT-LCD Panel Driving Section

Item 项目	Symbol 符号	Min. 最小值	Typ. 典型值	Max. 最大值	Unit 单位	Remark 备注
Power Supply Voltage	VDD	3.0	3.3	3.6	V	-
Power Supply Voltage	TP_VDD	2.8	3.3	3.3	V	-
Power Supply Current	I _{VDD}	-	50	-	mA	Note1
Power Supply Current	I _{TP_VDD}	-	10	-	mA	Note1
Logic Input High Voltage	V _{IH}	0.7VDD	-	VDD	V	-
Logic Input Low Voltage	V _{IL}	0	-	0.3VDD	V	-
Panel Power Consumption	P _{VDD}	-	0.198	-	Watt	Note1
Module Power Consumption	P _{LCM}	-	0.838	-	Watt	Note1,2

(Ta=+25°C, GND=0V)

Note1: Measurement Conditions (Video Mode): Full Screen Red Pattern, VDD=3.3V, 60Hz Refresh.

Note2: $P_{LCM} = P_{VDD} + P_{BL}$, About P_{BL} information, inference to 5.2 Back Light Driving Section.

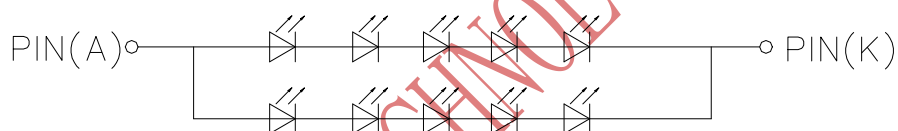
5.2 Back Light Driving Section

Item 项目	Symbol 符号	Min. 最小值	Typ. 典型值	Max. 最大值	Unit 单位	Remark 备注
Forward Voltage	V_F	-	16	17	V	Note1
Forward Current	I_F	-	40	-	mA	Note1
Backlight Power consumption	P_{BL}	-	0.64	-	Watt	Note1
LED life time	-	30000	-	-	Hrs	Note2
LED Quantity		10			PCS	

($T_a = +25^{\circ}\text{C}$, $GND = 0V$)

Note1: The LED driving condition is defined for each LED module

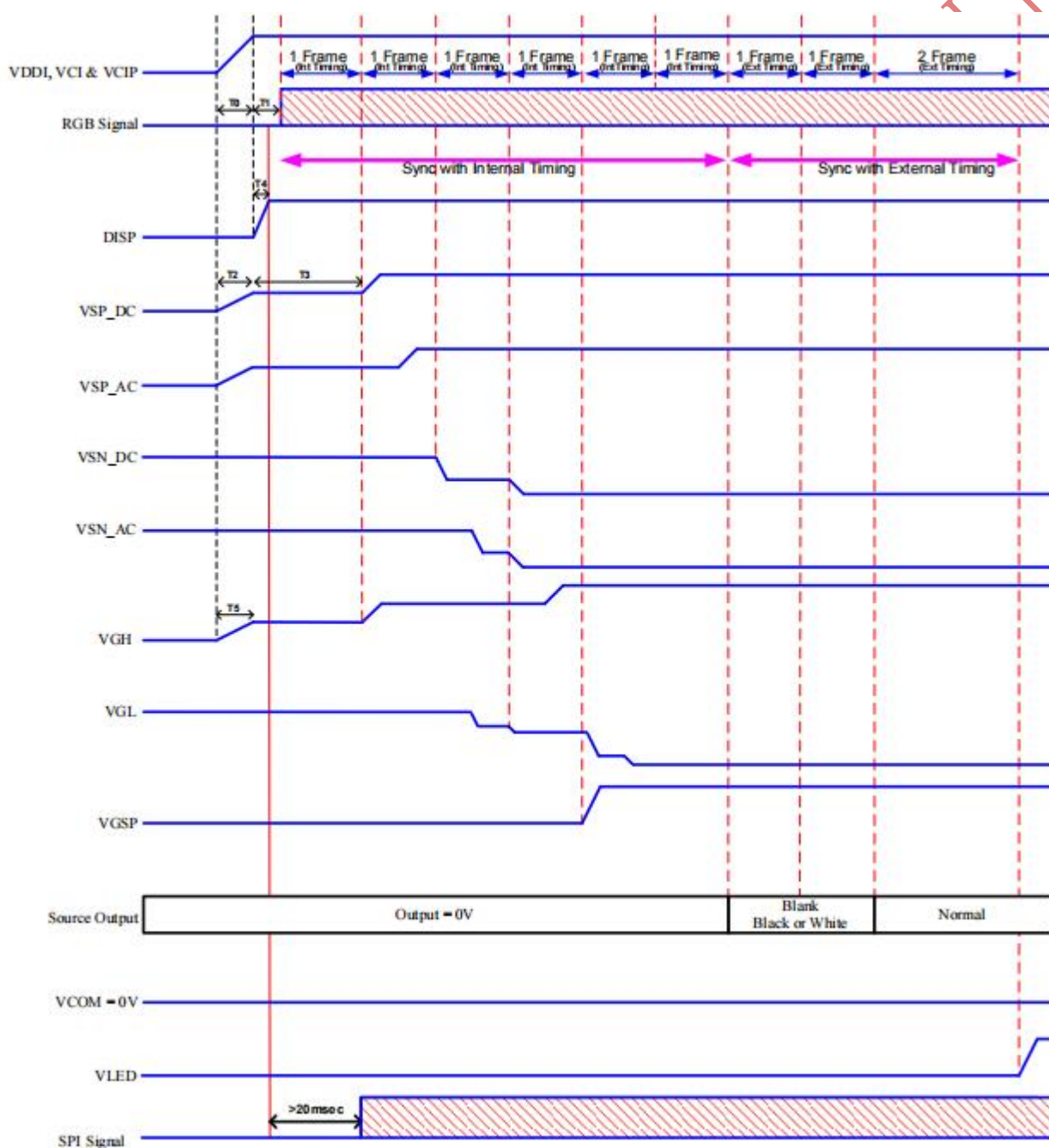
Note2: The "LED life time" is defined as the module brightness decrease to 50% of original brightness at $I_{LED} = 20\text{mA}$ (Per Led). The LED life time could be decreased if operating I_{LED} is larger than 20mA.



5.3 Power On/Off Sequence

5.3.1 LCM Power On/Off Sequence

Power On Sequence:

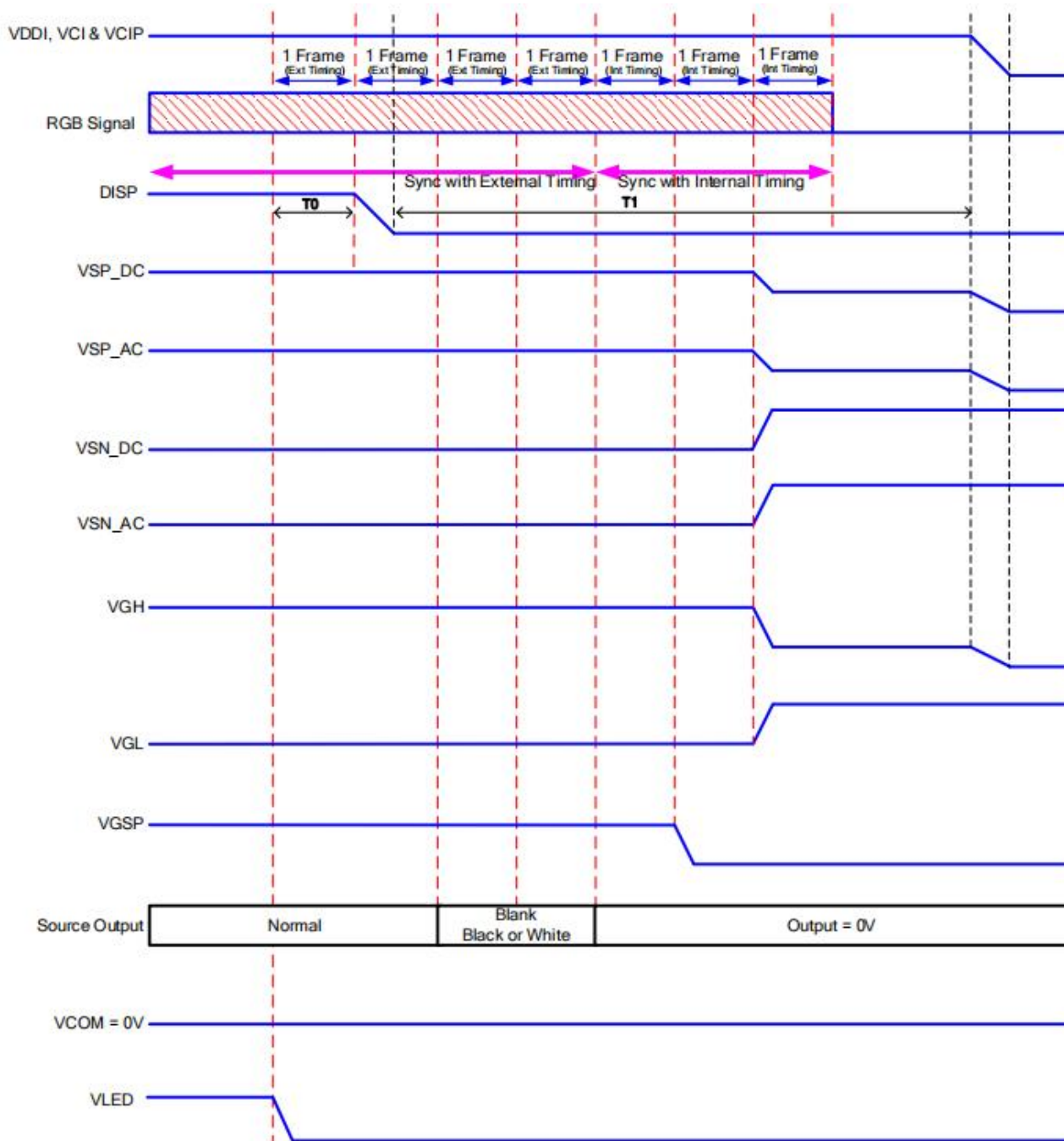


Item	Description	Min. Time
T0	Determined by the external power.	
T1	Time from stable VDDI, VCI, VCIP set-up to the first frame.	T1=0
T2	Time from VSP_DC=0V to VSP_DC=3.3V.	T2=T0
T3	Time from VSP_DC =3.3V to VSP_DC =6.0V.	T3=T1+ (1*Frame)
T4	Time from stable VDDI, VCI, VCIP set-up to DISP asserted.	T4=0
T5	Time from VGH=0V to VGH=3.3V.	T5=T0

Note: Recommend the LCM power on rise time T0= 0~ 1ms.

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Power Off Sequence:



Item	Description	Min. Time
T0	Time from backlight power off to DISP="L".	1*Frame
T1	Time from DISP="L" to LCM Power off.	5*Frame

5.3.2 CTP Power On/Off Sequence

The GPIO such as INT and I2C are advised to be low before powering on. Reset should be pulled down to be low before powering on. INT signal will be sent to the host after initializing all parameters and then start to report points to the host. If Power is down, the voltage of supply must be below 0.3V and Trst is more than 5ms.

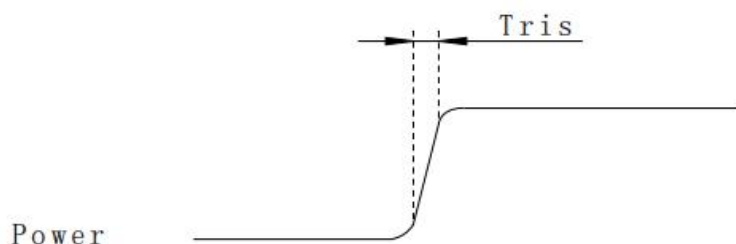


Figure 3-7 Power on time

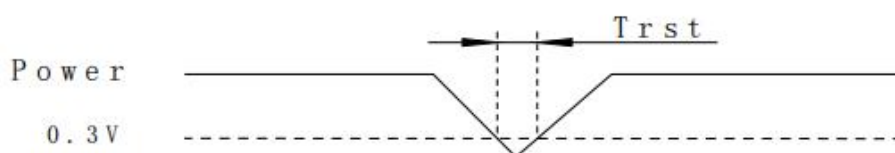


Figure 3-8 Power Cycle requirement

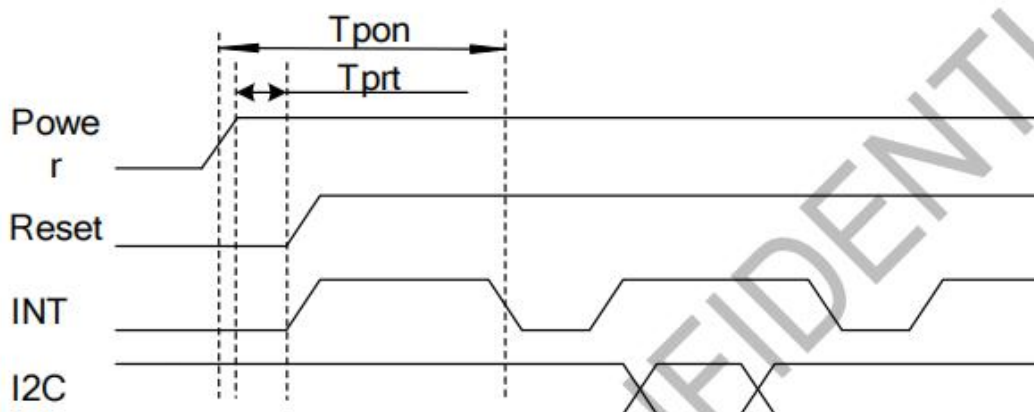


Figure 3-9 Power on Sequence

Reset time must be enough to guarantee reliable reset, the time of starting to report point after resetting approach to the time of starting to report point after powering on.

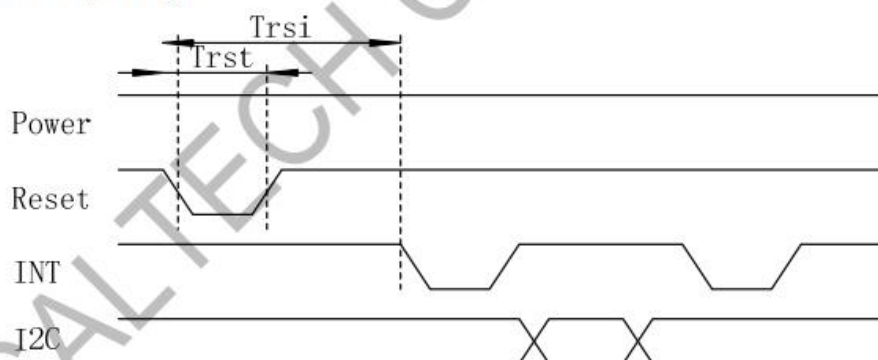


Figure 3-10 Reset Sequence

Table 3-5 Power on/Reset/Wake Sequence Parameters

Parameter	Description	Min	Max	Units
Tris	Rise time from 0.1VDD to 0.9VDD	-	3	ms
Tpon	Time of starting to report point after powering on	300	-	ms
Tprt	Time of being low after powering on	1	-	ms
Trsi	Time of starting to report point after resetting	300	-	ms
Trst	Reset time	5	-	ms

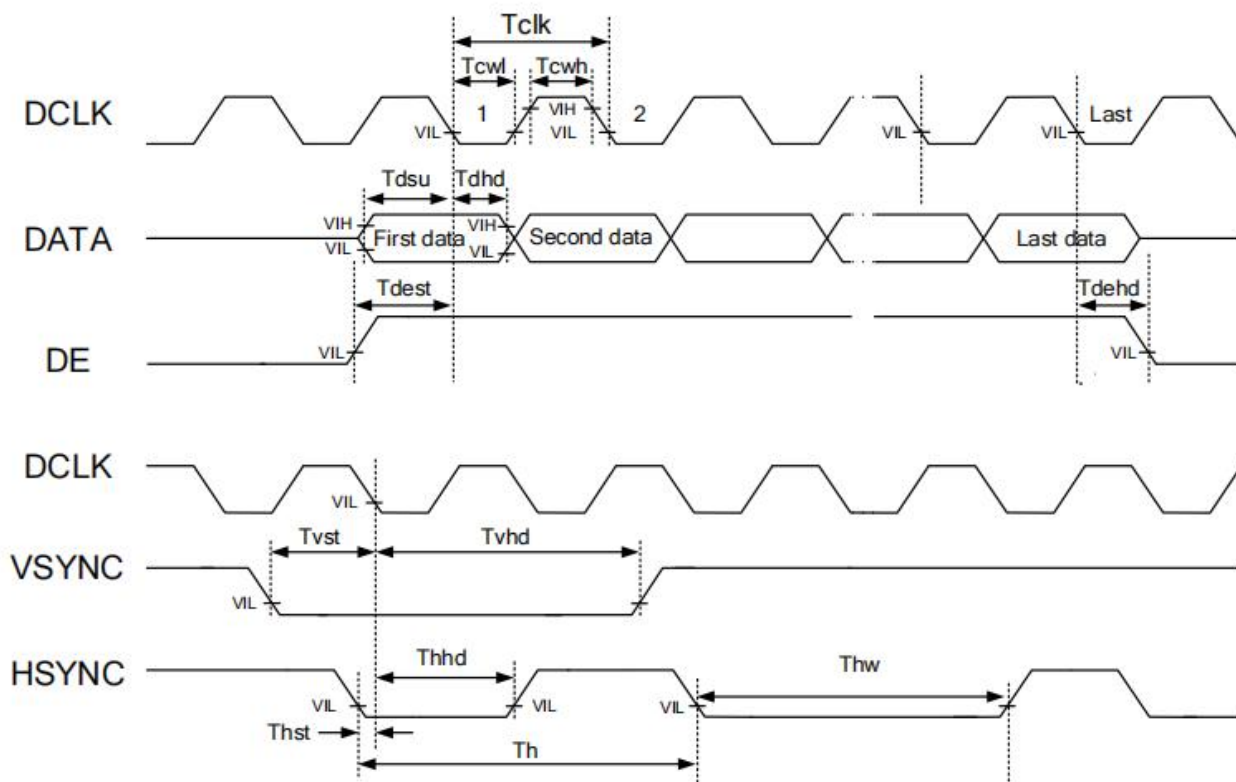
5.4 Timing Characteristics

AC Electrical Characteristics (VCIP=VCI=VDDI= 3.3V, VSSA= 0V, TA=25°C).

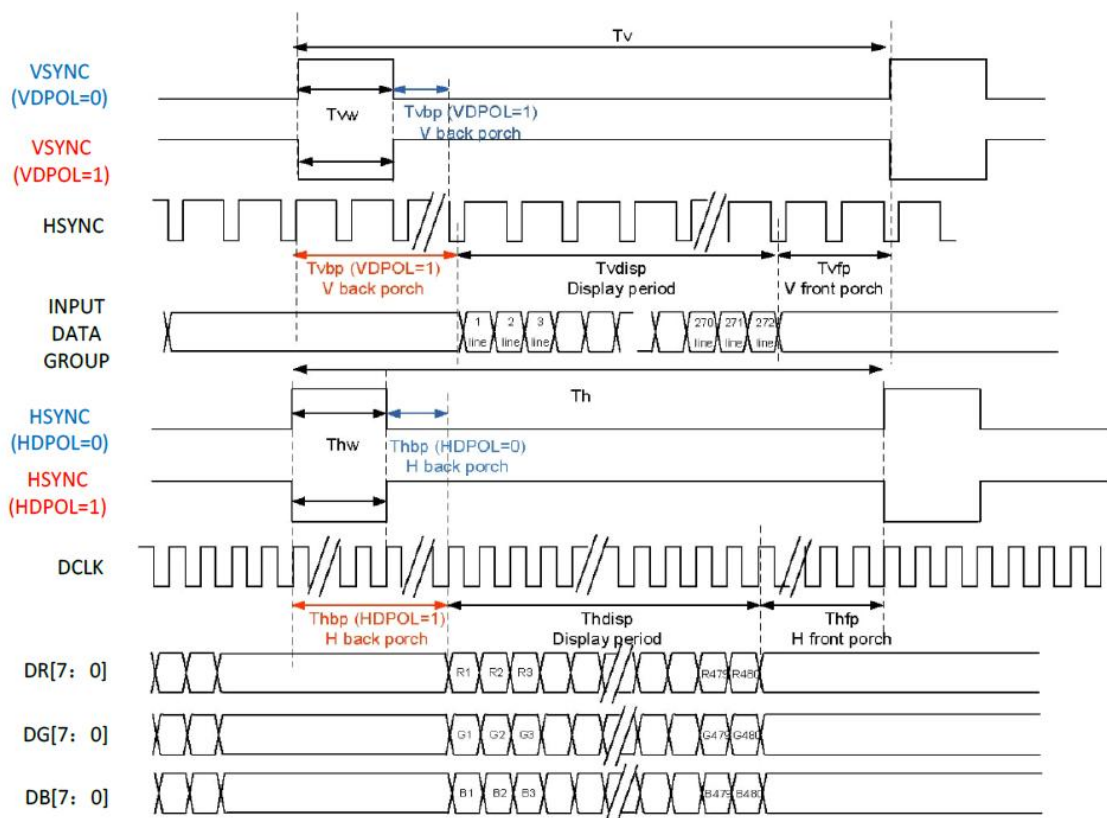
Item	Symbol	Min.	Typ.	Max.	Unit	Conditions
System operation timing						
VCI power source slew time	TPOR	-	-	20	ms	From 0V to 99% VCI
RESX pulse width	tRSTW	10	50	-	us	R=10Kohm, C=1uF
Input/ Output timing						
CLK pulse duty	Tcw	40	50	60	%	
Hsync width	Thw	2	-	-	DCLK	
Hsync period	Th	55	60	65	us	
Vsync setup time	Tvst	12	-	-	ns	
Vsync hold time	Tvhd	12	-	-	ns	
Hsync setup time	Thst	12	-	-	ns	
Hsync hold time	Thhd	12	-	-	ns	
Data setup time	Tdsu	12	-	-	ns	
Data hold time	Tdhd	12	-	-	ns	
DE setup time	Tdest	10	-	-	ns	
DE setup time	Tdehd	10	-	-	ns	
SD output stable time	Tst	-	-	12	us	Output settled within +20mV Loading = 6.8k+28.2pF.
GD output rise and fall time	Tgst	-	-	6	ns	Output settled (5%~95%), Loading = 4.7k+29.8pF
3-wire serial communication						
Delay between CSX and VSYNC	Tcv	1	-	-	us	
CSX input setup time	Ts0	50	-	-	ns	
Serial data input setup time	Ts1	50	-	-	ns	
CSX input hold time	Th0	50	-	-	ns	
Serial data input hold time	Th1	50	-	-	ns	
SCL pulse high width	Twh1	50	-	-	ns	
SCL pulse low width	Twl1	50	-	-	ns	
CSX pulse high width	Tw2	400	-	-	ns	

5.5 Timing Diagram

Clock and Data Input Timing Diagram



SYNC Mode Timing Diagram



Parallel 24-bit RGB Timing Table

480RGB X 272 Resolution Timing Table							
Item		Symbol	Min.	Typ.	Max.	Unit	Remark
DCLK Frequency		Fclk	8	9	12	MHz	
DCLK Period		Tclk	125	111	83	ns	
HSYNC	Period Time	Th	487	531	598	DCLK	
	Display Period	Thdisp	-	480	-	DCLK	
	Back Porch	Thbp	3	43	43	DCLK	By H_Blanking setting
	Front Porch	Thfp	4	8	75	DCLK	
	Pulse Width	Thw	2	4	75	DCLK	
VSYNC	Period Time	Tv	276	292	321	H	
	Display Period	Tvdisp	-	272	-	H	
	Back Porch	Tvbp	2	12	12	H	By V_Blanking setting
	Front Porch	Tvfp	2	8	37	H	
	Pulse Width	Tvw	2	4	37	H	

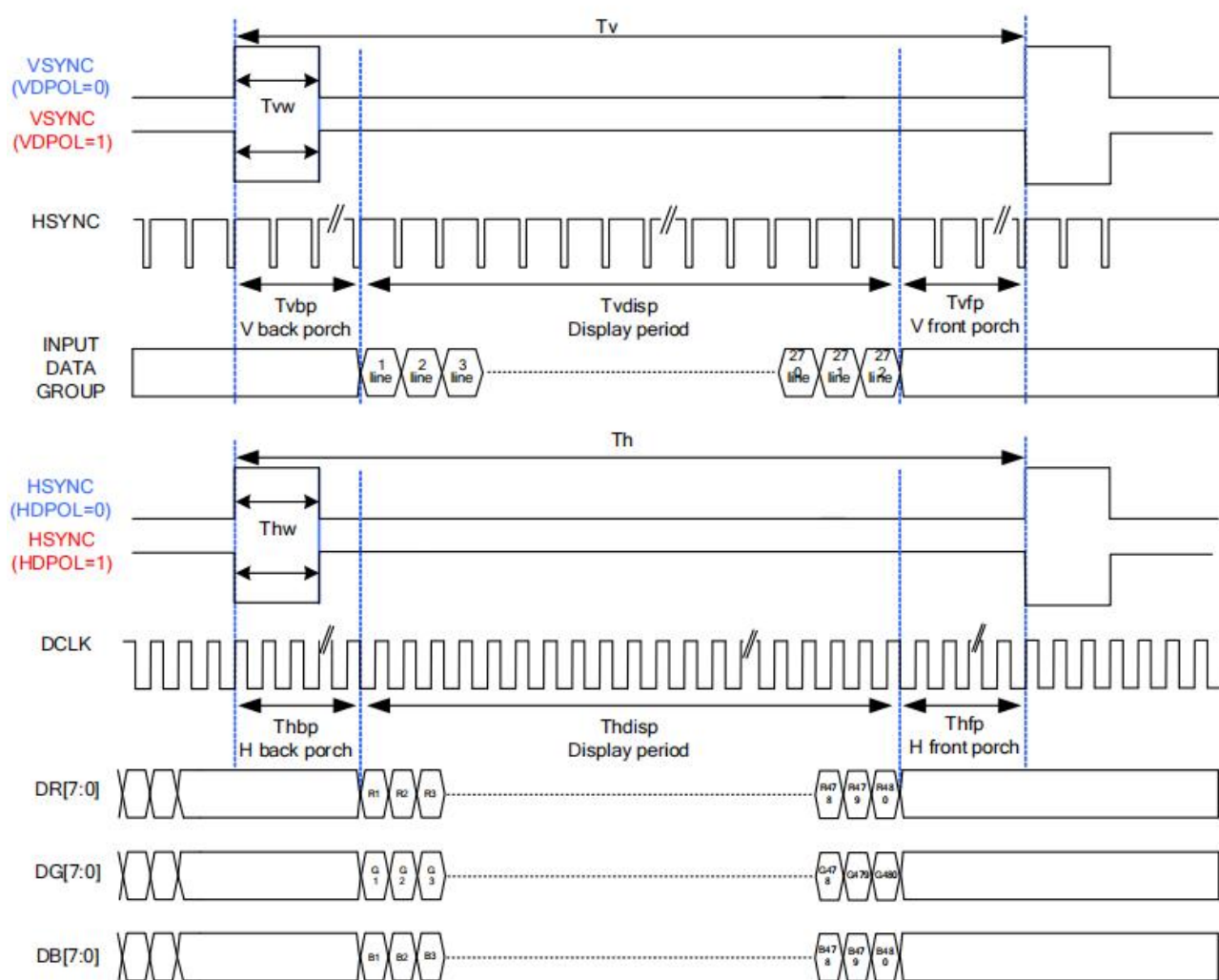
Note: 1.It is necessary to keep Tvbp =12 and Thbp =43 in sync mode. DE mode is unnecessary to keep it.

2.Thbp+Thfp >=7

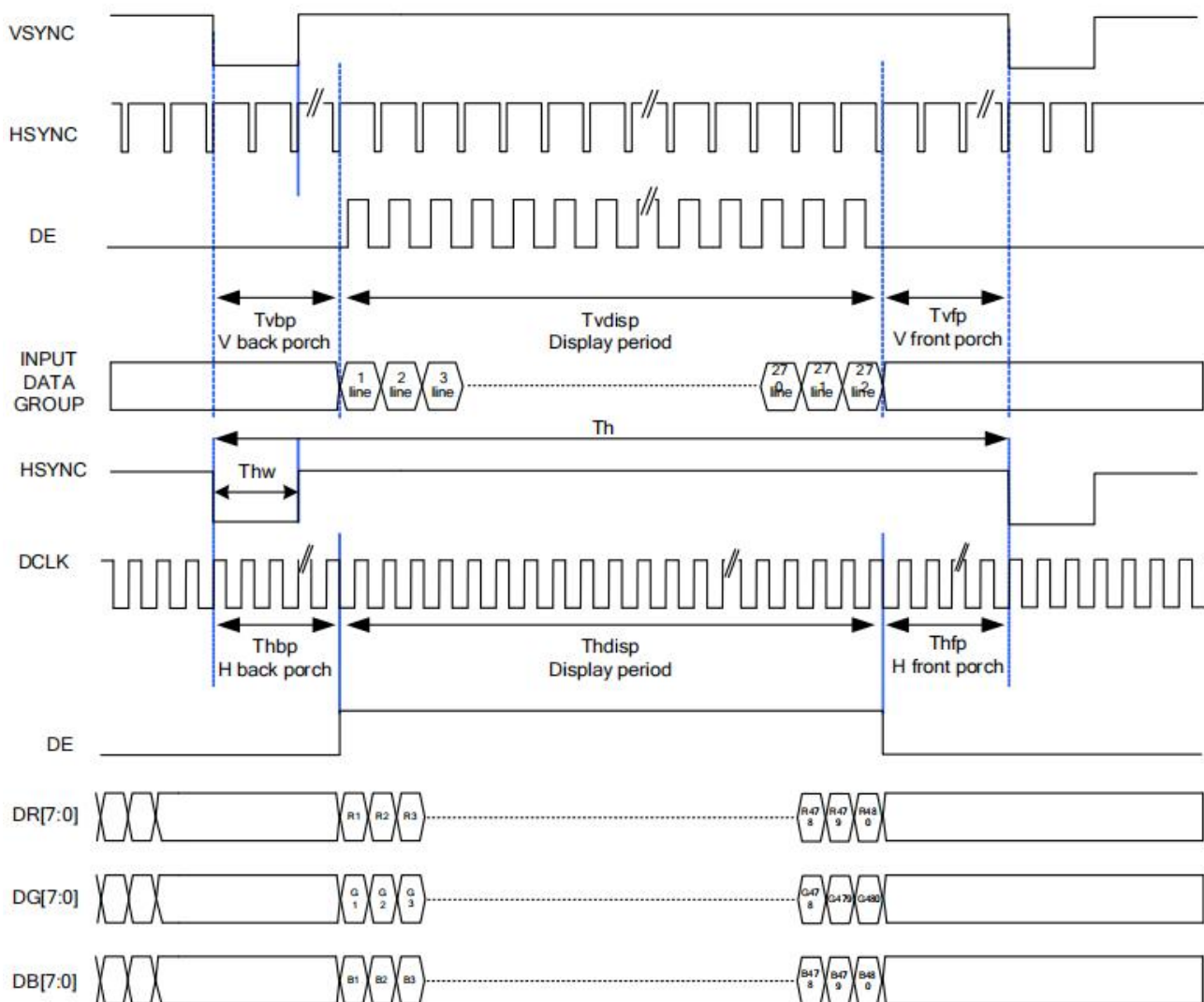
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SYNC-DE Mode Timing Diagram

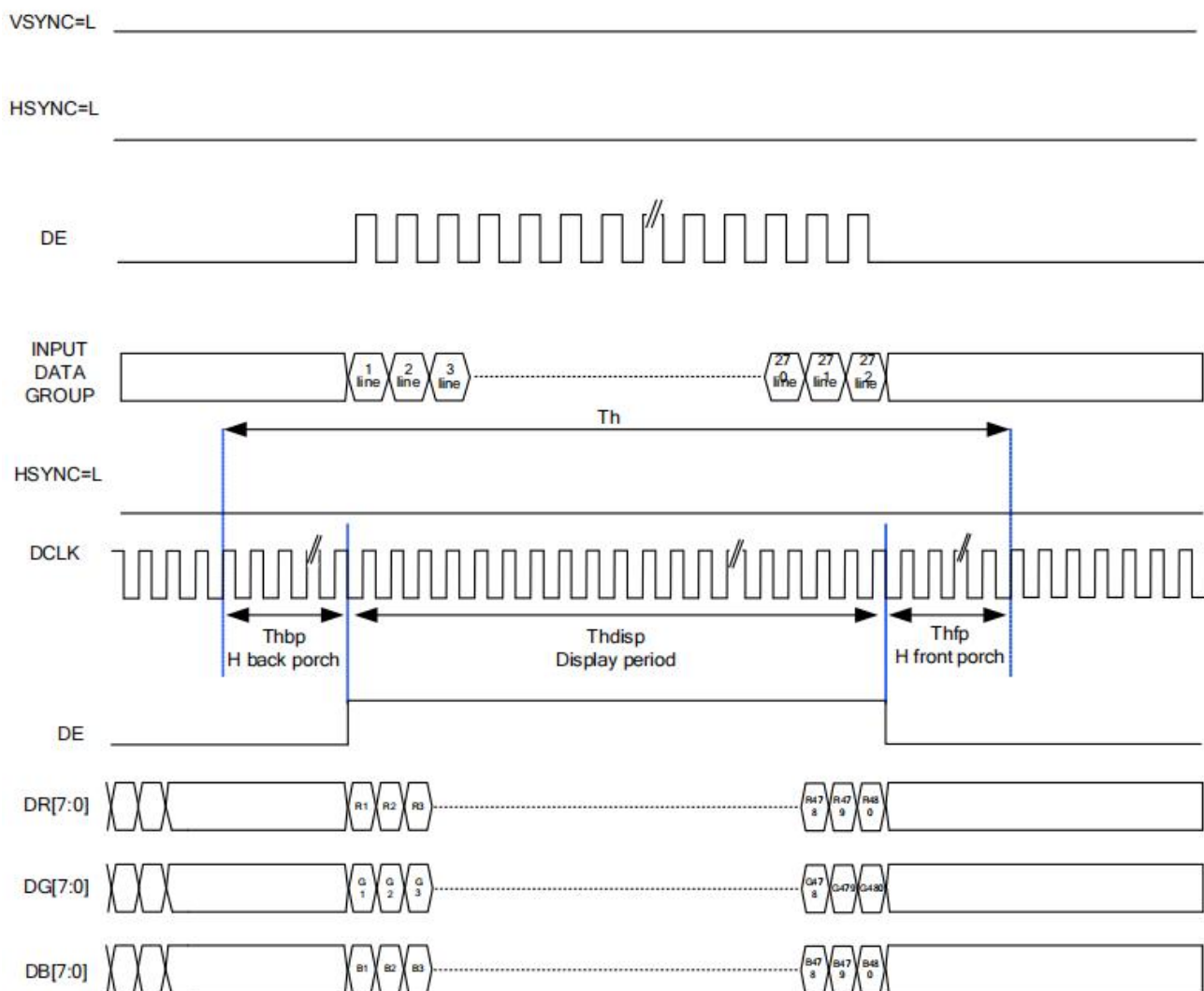
SYNC Mode Timing Diagram



SYNC-DE Mode Timing Diagram



DE Mode Timing Diagram



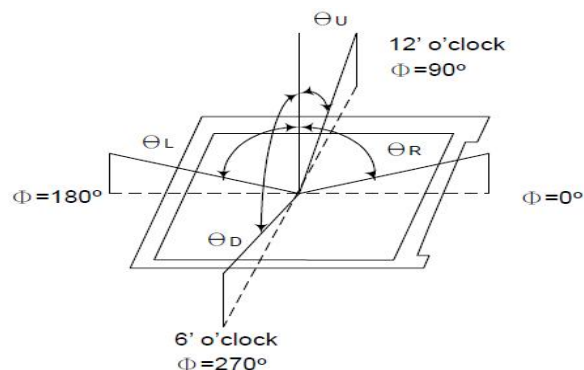
6. OPTICAL CHARACTERISTICS

Parameter 参数	Symbol 符号	Condition 条件	Min. 最小值	Typ. 典型值	Max. 最大值	Unit 单位	Remark 备注
Contrast Ratio	C/R	$\theta = 0^\circ$	300	500	-	-	Note(4)
NTSC Ratio	S	$\theta = 0^\circ$	-	52	-	%	Note(7)
Luminance	L	$\theta = 0^\circ$	200	250	-	cd/m ²	Note(5)
Luminance uniformity	U _w	$\theta = 0^\circ$	70	80	-	%	Note(3)
Response Time	T _R + T _F	25 °C	-	25	50	ms	Note(2)
Color Coordination	W _X	$\theta = 0^\circ$ (Center) Normal viewing angle B/L On	-0.04	0.28	+0.04	NTSC (x,y)	Note(6)
	W _Y			0.30			
	R _X			0.618			
	R _Y			0.343			
	G _X			0.306			
	G _Y			0.527			
	B _X			0.139			
	B _Y			0.098			
Viewing Angle	θ_L	C/R>10	-	70	-	Degree	Note(1)
	θ_R		-	70	-		
	θ_U		-	50	-		
	θ_D		-	60	-		

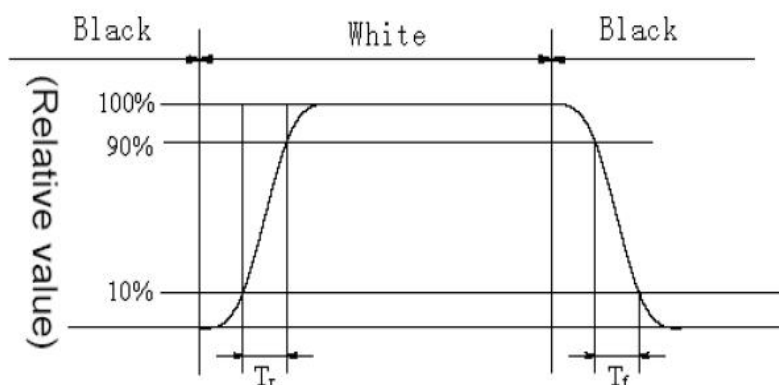
Test Conditions:

1. the ambient temperature is +25°C.
2. The test systems refer to Note 8.

Note1: Definition of Viewing Angle: The viewing angle range that the CR>10

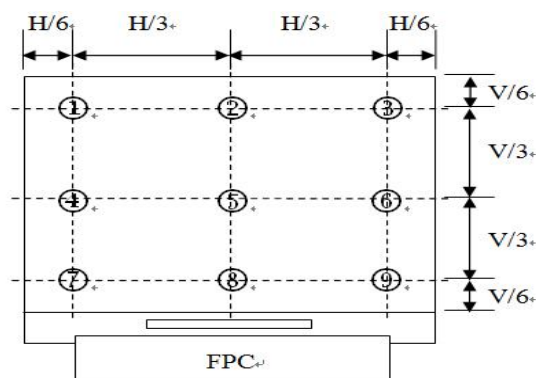


Note2: Definition of Response time: Sum of T_R and T_F



Note 3: Definition of Luminance Uniformity: Active area is divided into 9 measuring areas, every measuring point is placed at the center of each measuring area.

$$\text{Luminance Uniformity} = \frac{\text{Min Luminance of white among 9-points}}{\text{Max Luminance of white among 9-points}} \times 100\%$$



Note4: Definition of Contrast Ratio (CR): measured at the center point of panel

$$\text{Contrast ratio (CR)} = \frac{\text{Luminance measured when LCD on the "White" state}}{\text{Luminance measured when LCD on the "Black" state}}$$

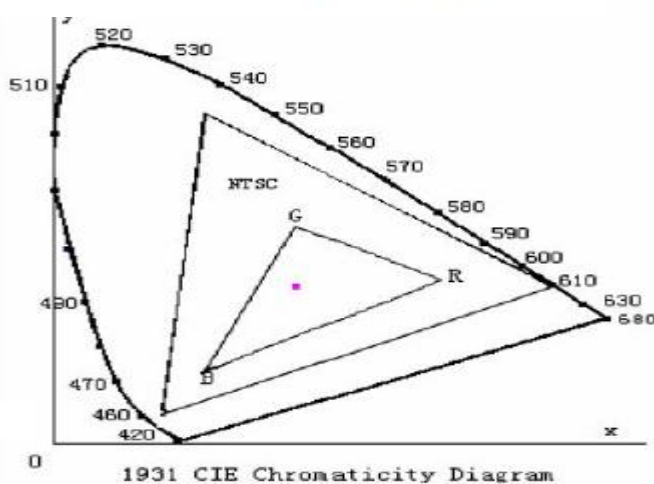
Note 5: Definition of Luminance: Center Luminance of white is defined as luminance values of 1point average across the LCD surface.

Note 6: Definition of Color Chromaticity (CIE 1931)

Color coordinates of white & red, green, blue measured at center point of LCD.

Note 7: Definition of NTSC ratio:

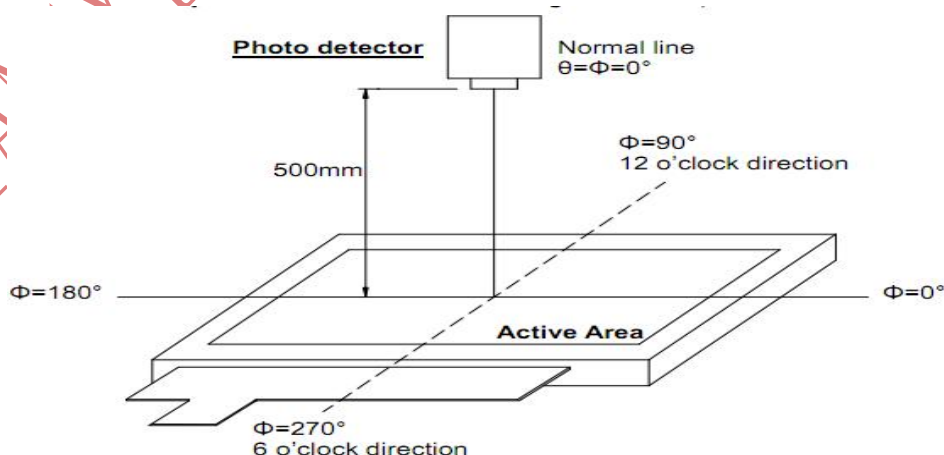
$$\text{NTSC ratio} = \frac{\text{Area of RGB triangle}}{\text{Area of NTSC triangle}}$$



Note 8: Definition of measurement system.

optical

The optical characteristics should be measured in dark room. After 5 minutes operation, the optical properties are measured at the center point of the LCD screen. (Response time is measured by Photo detector TOPCON BM-7, Field of view: 1°/Height: 500mm.)



7. RELIABILITY

Item 项目	Test Condition 测试条件	Remark 备注
High Temperature Storage	Ta =+80°C / 96Hours	Note1,2,3
Low Temperature Storage	Ta =-30°C / 96Hours	Note1,2,3
High Temperature Operating	Ta =+70°C / 96Hours	Note1,2,3
Low Temperature Operating	Ta =-20°C / 96Hours	Note1,2,3
Temperature Cycle storage Test	-30°C/30min ↔+80°C /30min for 30cycles,Transfer time less than 5min	Note2,3
Thermal humidity storage Test	60°C x 90%RH / 96Hours	Note2,3
Package Vibration Test	Frequency: 10Hz~55Hz,Amplitude:1.5mm, 1 hrs for each direction of X, Y, Z	Note2
Packing shock test	Drop to the ground from 60cm height, 1 corner, 3 edges, 6 surfaces.	Note2

Inspection after Test:

Note1:Ta is the ambient temperature of samples.

Note 2: In the standard condition, there shall be no practical problem that may affect the display function. After the reliability test, the product only guarantees operation, but doesn't guarantee all the cosmetic specification.

Note 3: Before cosmetic and function tests, the product must have enough recovery time, at least 2 hours at room temperature.

Note 4: In case of malfunction defect caused by ESD damage, if it would be recovered to normal state after resetting, it would be judged as a good part.

8. PACKAGE DRAWING

